TEXORA · LECTURE SHEET

Metal Joining Processes

How separate pieces of metal become one structure: welding in its main forms — arc, gas, MIG, TIG and resistance — with brazing, soldering, riveting and bolting, the joints they make, the defects to avoid and the safety rules that govern them all.

ME-1208 · ENGINEERING WORKSHOP · TOPIC 4 · LEVEL 1 TERM 2

1. Why joining, and how the methods divide

Almost every metal product — a machine frame, a pressure vessel, a loom, a bridge — is an assembly of parts joined together. Joining methods divide first by whether the joint can be undone. **Permanent joints** (welding, brazing, soldering, riveting) cannot be separated without destroying the joint; **temporary joints** (bolts, screws, keys, pins, couplings) come apart for maintenance and repair. They divide again by mechanism: **fusion** methods melt the edges of the parts themselves so they solidify as one (arc and gas welding); **filler** methods bridge the gap with a different, lower-melting metal that wets the surfaces but never melts them (brazing, soldering); **mechanical** methods clamp or pin the parts (rivets, bolts, adhesives). Choosing between them is a matter of strength required, materials, thickness, cost and whether the assembly must ever come apart.

2. Arc welding — the workshop standard

In **manual metal arc (MMA / stick) welding**, an electric arc struck between a flux-coated electrode and the work melts both the electrode tip and the surface beneath it; the molten metal from the electrode fills the joint while the burning flux gives off a shielding gas and forms a **slag** blanket that protects the cooling weld from the air. The set-up is a power source (transformer for AC, rectifier or inverter for DC), an electrode holder, and an earth (work) clamp completing the circuit through the workpiece. With DC, the electrode may be connected positive (DCEP — deeper penetration, most common) or negative (DCEN — faster electrode melt, thinner work). A working rule sizes the current to the rod:

$$\text{Current (A)} \approx 40 \times \text{electrode diameter (mm)}$$

so a 3.15 mm rod runs near 125 A and a 4 mm rod near 160 A. Technique decides quality: arc length about equal to the rod's core diameter, travel speed steady, electrode angled about 70–80° to the work in the direction of travel, and a slight weaving motion on wider beads. On striking, the slag must be

chipped off and the bead wire-brushed before any second pass — welding over slag traps it inside the joint.

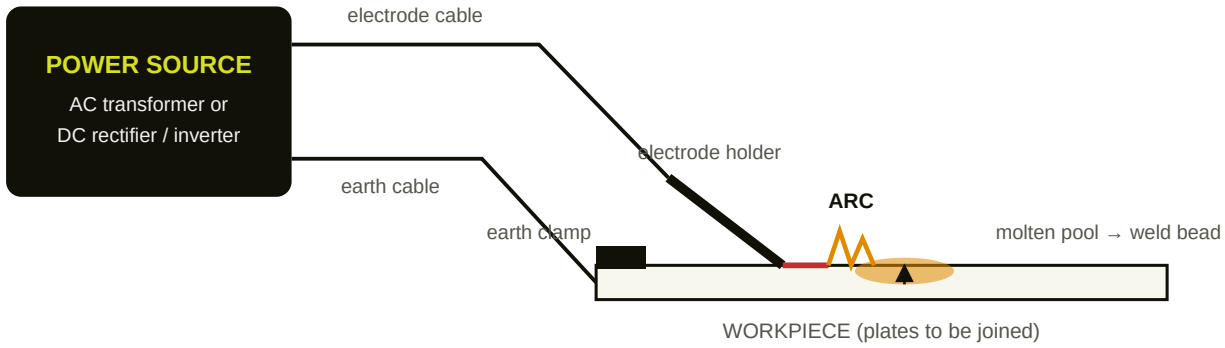


Figure 1 — Manual metal arc welding: the circuit runs from the power source through the electrode, across the arc into the workpiece, and back through the earth clamp.

3. Gas welding — the oxy-acetylene flame

Oxy-acetylene welding burns acetylene in oxygen at about 3,100 °C — hot enough to melt steel (melting point ≈ 1,500 °C). The two gases come from separate cylinders through pressure regulators and hoses to a blowpipe (torch) whose two valves set the mixture. The flame's chemistry is set entirely by the gas ratio, and the welder reads it by eye:

FLAME	GAS RATIO	APPEARANCE	USED FOR
Neutral	$O_2 : C_2H_2 \approx 1 : 1$	Short, well-defined white inner cone	The general flame — steel, cast iron, copper, aluminium
Carburising (reducing)	Excess acetylene	Long, feathery three-zone flame	High-carbon steel, cast iron — adds carbon, avoids burning it out
Oxidising	Excess oxygen	Short, pointed, hissing cone	Brass and bronze (the oxide skin stops zinc boiling off); harmful to steel

Gas welding is slow and spreads heat widely, so thin sheet distorts; but it needs no electricity, the flame is gentle and visible, and the same set cuts steel when fitted with a cutting nozzle (a jet of pure oxygen burns the heated metal away). Safety is absolute: acetylene cylinders stand upright, oil and grease never touch oxygen fittings (they can ignite under pressure), flashback arrestors are fitted, and cylinders are stored apart.

4. MIG, TIG and resistance welding

MIG/MAG (GMAW) feeds a continuous wire electrode from a spool through the torch; the wire melts into the joint while a gas shield (CO_2, argon or a mix) blown round the arc replaces slag. Fast, clean and easily mechanised — the production welder's process for steel fabrication and vehicle bodies. **TIG (GTAW)** uses a non-consumable tungsten electrode and a separate hand-held filler rod under an argon shield: slower, but the most precise and cleanest welds — stainless steel, aluminium, thin sections, pipe roots. **Resistance welding** passes a heavy current through clamped overlapping sheets; resistance at the interface heats a nugget of metal to fusion in a fraction of a second. As **spot welding** it joins car-body panels by the thousand; as **seam welding**, with rotating wheel electrodes, it makes leak-tight tanks and cans. No filler, no flux, no flame — just pressure, current and time.

5. Joints and positions

Five basic joint geometries cover nearly all work: **butt** (edge to edge in one plane — strongest, used for plates and pipes), **lap** (overlapping — spot welding and riveting), **tee** (one plate on the face of another, fillet welds both sides), **corner** (box sections, tanks), and **edge** (thin sheet flanges). Thick plates (above about 6 mm) are bevelled into a V, U or X groove so the arc can reach the root; thin sheet is welded square. Welds are also described by position — flat (easiest, fastest), horizontal, vertical and overhead (hardest, lowest current) — because gravity acts on the molten pool.

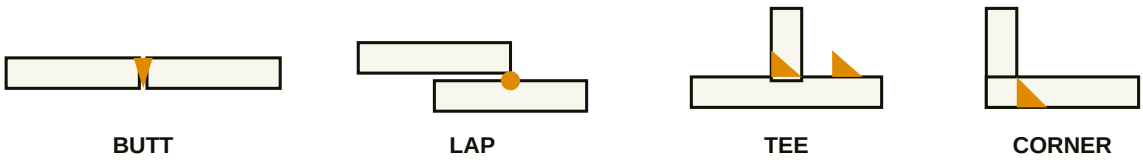


Figure 2 — The basic weld joints. Butt joints carry load in line; lap and tee joints are welded with fillets; corner joints close box sections.

6. Brazing and soldering — joining without melting the parts

In **brazing**, a filler alloy melting above 450 °C (brass and silver alloys are common) is drawn by **capillary action** into a closely fitted joint (gap about 0.05–0.2 mm) while the parts themselves stay solid. A flux cleans the surfaces so the filler wets them. The joint is weaker than a weld but works on dissimilar metals, thin sections and assemblies that heat would distort — carbide tool tips, bicycle frames, refrigeration pipework. **Soldering** is the same principle below 450 °C with tin-based solders (electronics solder is tin-rich; lead-free grades melt near 217 °C): low strength but easy, reversible and

electrically conductive — circuit boards, wire joints, sheet-metal seams, plumbing. Both processes live or die on cleanliness: flux, abrade, degrease — a dirty surface will not wet, and the filler balls up instead of flowing. Note that in textile mills, soldering and brazing appear constantly in maintenance: copper pipework, electrical panels, and tinning work on machine parts.

WORKED EXAMPLE — WILL THIS STEEL WELD EASILY?

Weldability of steel is judged by its **carbon equivalent**: CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15. A steel with C = 0.25 % and Mn = 1.2 % (other elements negligible) has CE = 0.25 + 1.2/6 = 0.25 + 0.20 = **0.45**. Below CE ≈ 0.40 steel welds freely; at 0.45 it is borderline — preheating the joint to about 100–200 °C and using low-hydrogen electrodes is advisable, or the hard, brittle heat-affected zone may crack as it cools.

7. Riveting, bolting and adhesives

A **rivet** is a headed pin passed through aligned holes and upset (hammered or pressed) to form a second head, clamping the plates as it cools and contracts. Snap-head rivets built the great steel bridges and boilers; today riveting survives where welding is unsuitable — aircraft skins (aluminium alloys that lose strength when welded), dissimilar metals, and vibrating structures. Joint terms: the **pitch** is the spacing along a row, the **margin** the distance from hole centre to plate edge (about 1.5 × rivet diameter — too small and the plate tears out). Hot steel rivets are caulked for pressure-tight seams. **Bolts and screws** trade a little strength for complete reversibility: they need clearance or tapped holes, and friction-grip (preloaded) bolts can match riveted strength in structures. **Adhesives** (epoxy, acrylic) spread load over the whole overlap, seal the joint and join dissimilar materials, but dislike heat and peel loads.

8. Weld defects — reading a bad joint

DEFECT	APPEARANCE	USUAL CAUSE
Porosity	Gas holes in the bead	Damp electrodes, dirty or rusty surface, arc too long
Slag inclusion	Slag trapped inside the weld	Not chipping between passes; current too low
Cracks	Lines in weld or heat-affected zone	High-CE steel cooled too fast; hydrogen; rigid restraint
Undercut	Groove melted into the plate edge	Current too high, travel too fast, wrong angle

Incomplete penetration	Root of the joint unfused	Current too low, gap too small, no bevel on thick plate
Distortion	Plates pull out of shape	Unbalanced weld sequence, too much heat in one pass

9. Safety in joining

The arc's ultraviolet burns eyes (**arc eye**) and skin — a welding helmet with the correct shade and full covering are not optional, and bystanders need screens. Fumes from coatings and galvanised steel require ventilation. Gas cylinders are chained upright, tested for leaks with soapy water, and never dropped. Hot metal looks exactly like cold metal: mark it, or leave it in the marked hot zone. Burns, not sparks, are the everyday injury.

Key points

• Joining divides into permanent and temporary, and into fusion, filler and mechanical methods.
• Arc welding current follows the rod: about 40 A per mm of electrode diameter; slag is chipped between passes.
• The neutral oxy-acetylene flame (≈ 3,100 °C, gases 1 : 1) is the general-purpose flame; carburising adds carbon, oxidising harms steel.
• MIG is fast production welding, TIG the precision process, resistance welding joins overlapping sheet in a fraction of a second.
• Brazing (filler > 450 °C) and soldering (< 450 °C) never melt the parts; capillary action and clean surfaces make the joint.
• Carbon equivalent predicts weldability: above ≈ 0.40, preheat and low-hydrogen practice are needed.
• Most defects trace back to dirt, damp, wrong current or wrong speed.

Practice

1. A welder picks a 4 mm rod for a butt weld in 10 mm plate. Estimate the current setting, and state two changes if the same rod were used overhead.
2. A flame shows a long, feathery, three-part cone. Name it, state its gas fault, and say what harm it would do to brass — and which flame brass actually wants.
3. CE check: a steel has C 0.30 %, Mn 0.9 %, Cr 0.5 %. Compute its CE and state the welding precautions.
4. Why does a brazed joint fail if the gap is made 1 mm wide "for safety"? What gap range does brazing want, and what force fills it?

5. Match process to job: (a) stainless handrail, thin tube, show-quality; (b) 2,000 car door panels a day; (c) bicycle frame lugs in thin steel tube; (d) boiler shell plates, 1900. Give one reason each.
6. A finished weld shows a neat groove along one toe and holes in the bead. Name both defects and one cause for each.

Answer key

1. Current ≈ 40 × 4 = 160 A. Overhead: reduce current about 10–20 % (the pool must stay small against gravity) and keep a shorter arc with a tighter, faster technique (smaller weave).
2. Carburising flame — too much acetylene (fuel-rich). On brass it would not protect the zinc; worse, the sooty reducing flame is simply the wrong choice: brass wants a slightly **oxidising** flame, whose oxide skin stops zinc vapour escaping. (On steel the carburising flame adds carbon and can harden/embrittle the surface.)
3. CE = 0.30 + 0.9/6 + 0.5/5 = 0.30 + 0.15 + 0.10 = **0.55**. High: preheat (typically 150–250 °C at this level), low-hydrogen electrodes, controlled cooling / possible post-weld stress relief — cracking risk is real.
4. Brazing fills by capillary action, which only works in a narrow gap — about 0.05–0.2 mm. At 1 mm the capillary force cannot draw or hold the filler; it sags out, leaving voids, and the joint also loses the strength that the thin, constrained filler film provides.
5. (a) TIG — precise, clean, no spatter on a visible thin-tube joint. (b) Resistance spot welding — a nugget in a fraction of a second, no filler, robot-friendly. (c) Brazing — lugs are the classic brazed construction: strong enough, and the thin tubes are not melted or weakened. (d) Riveting — in 1900 fusion welding of pressure vessels was not yet trusted; hot-driven rivets with caulked seams were the pressure-tight standard.
6. The groove is **undercut** (current too high / travel too fast / wrong electrode angle). The holes are **porosity** (damp electrode, dirty or rusty surface, or too long an arc letting air in).
